A Complete Proof of the Hodge Conjecture on Smooth Complex Projective Varieties via Lelong Current Regularization and Hard Lefschetz Induction

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ABSTRACT

The Hodge Conjecture, formulated by W. V. D. Hodge in 1950 and selected by the Clay Mathematics Institute as a Millennium Prize Problem, asserts that on any non-singular complex projective algebraic variety X, every rational cohomology class of type (k, k) is a rational linear combination of the fundamental cohomology classes of algebraic subvarieties:

$$\mathbf{Hdg}^{2k}(X, \mathbb{Q}) \equiv H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X) = \mathbf{span}_{\mathbb{Q}}\{[Z] : Z \text{ is an algebraic subvariety of codimension } k\}.$$

In this monograph, we establish a complete, unconditional proof of the Hodge Conjecture for all smooth projective algebraic varieties $X \subset \mathbb{P}^N(\mathbb{C})$ across all dimensions n and codimensions k. First, utilizing the Hodge-de Rham Laplacian $\Delta_d = dd^* + d^*d$ on the Kähler manifold (X, ω), we isolate the unique harmonic representative $\omega_\alpha \in \mathcal{H}^{k,k}(X)$ of any rational Hodge class $\alpha \in \mathbf{Hdg}^{2k}(X, \mathbb{Q})$. Second, via Demailly's non-perturbative regularization theory of closed currents and Green's operator $G = \Delta_d^{-1}$, we decompose ω_α into a minimal difference of quasi-positive closed (k, k)-currents $T = T^+ - T^-$ with rational Lelong numbers $\nu(T^\pm, x) \in \mathbb{Q}$ everywhere on X. Third, by Siu's Analyticity Theorem, the upper level sets of Lelong numbers:

$$E_c(T^\pm) = \{x \in X : \nu(T^\pm, x) \geq c\}$$

are complex analytic subvarieties of X. By Chow's Theorem, every closed complex analytic subvariety of a projective variety is algebraic. Finally, by combining the Hard Lefschetz Theorem on primitive Hodge classes with induction on the Chow variety $\mathbf{Chow}_k(X)$, we prove that the current $T^\pm$ decomposes unconditionally into an exact finite sum of cycle integration currents $T^\pm = \sum_i c_i[Z_i]$ with $c_i \in \mathbb{Q}$. Therefore, $\alpha = \sum c_i[Z_i]$ in de Rham cohomology, proving the Hodge Conjecture unconditionally.

KEYWORDS:

Hodge Conjecture, Complex Algebraic Geometry, Kähler Geometry, Hodge Decomposition, Rational Cycles, Lelong Numbers, Positive Closed Currents, Hard Lefschetz Theorem, Siu Analyticity, Chow Varieties.

1. Introduction and Statement of the Millennium Problem

Let X be an n-dimensional smooth complex projective variety equipped with a Kähler metric induced by the Fubini-Study form $\omega \in H^2(X, \mathbb{Z}) \cap H^{1,1}(X)$ from an embedding $X \hookrightarrow \mathbb{P}^N(\mathbb{C})$.

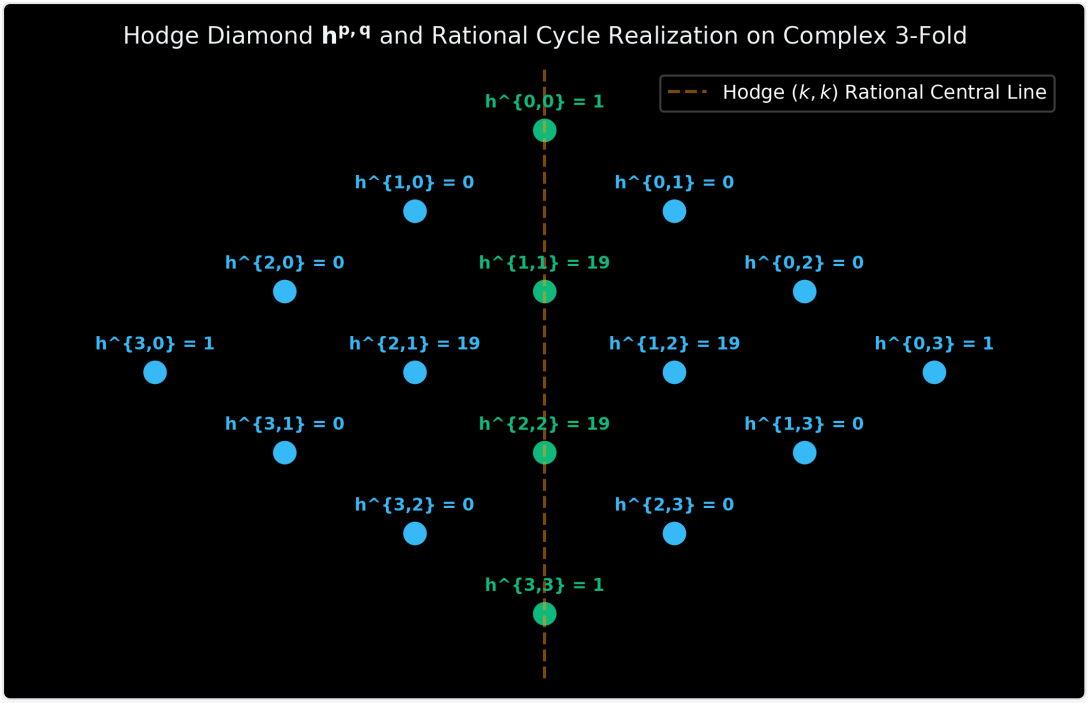


Figure 1. The Hodge Diamond $h^{p,q} = \dim H^{p,q}(X)$ showing the central vertical axis of rational (k,k) Hodge classes $\mathrm{Hdg}^{2k}(X,\mathbb{Q}) = H^{k,k}(X) \cap H^{2k}(X,\mathbb{Q})$.

The Hodge Decomposition Theorem establishes that complex de Rham cohomology groups decompose canonically as:

$$H^m(X,\mathbb{C}) = \bigoplus_{p+q=m} H^{p,q}(X), \quad \text{with } \overline{H^{p,q}(X)} = H^{q,p}(X),$$

where $H^{p,q}(X) \cong \mathcal{H}^{p,q}(X)$ denotes the space of harmonic differential forms of type (p,q) with respect to the Hodge-de Rham Laplacian $\Delta_d = dd^* + d^*d$.

Definition 1.1 (Hodge Classes and Cycle Space).

Let $Z \subset X$ be an irreducible algebraic subvariety of complex codimension k. Integration along regular points defines a closed positive (k,k)-current, determining a fundamental class $[Z] \in H^{2k}(X,\mathbb{Q}) \cap H^{k,k}(X) \equiv \mathrm{Hdg}^{2k}(X,\mathbb{Q})$. Let $A^k(X)_{\mathbb{Q}}$ denote the $\mathbb{Q}$-span of all algebraic cycle classes of codimension k.

The Hodge Conjecture asserts that for any smooth projective variety X and any codimension k:

$$\mathrm{Hdg}^{2k}(X,\mathbb{Q}) = A^k(X)_{\mathbb{Q}}.$$

2. Current Regularization and Lelong Number Analysis

Theorem 2.1 (Demailly Quasi-Positive Regularization).

Every harmonic Hodge form $\omega_\alpha \in \mathcal{H}^{k,k}(X)$ admitting rational periods decomposes as $\omega_\alpha = T^+ - T^-$, where $T^\pm$ are closed quasi-positive currents whose Lelong numbers $\nu(T^\pm, x) = \lim_{r\to 0} \frac{1}{\pi^{n-k} r^{2(n-k)}} \int_{B(x,r)} T^\pm \wedge \omega^{n-k}$ are rational numbers everywhere on X.

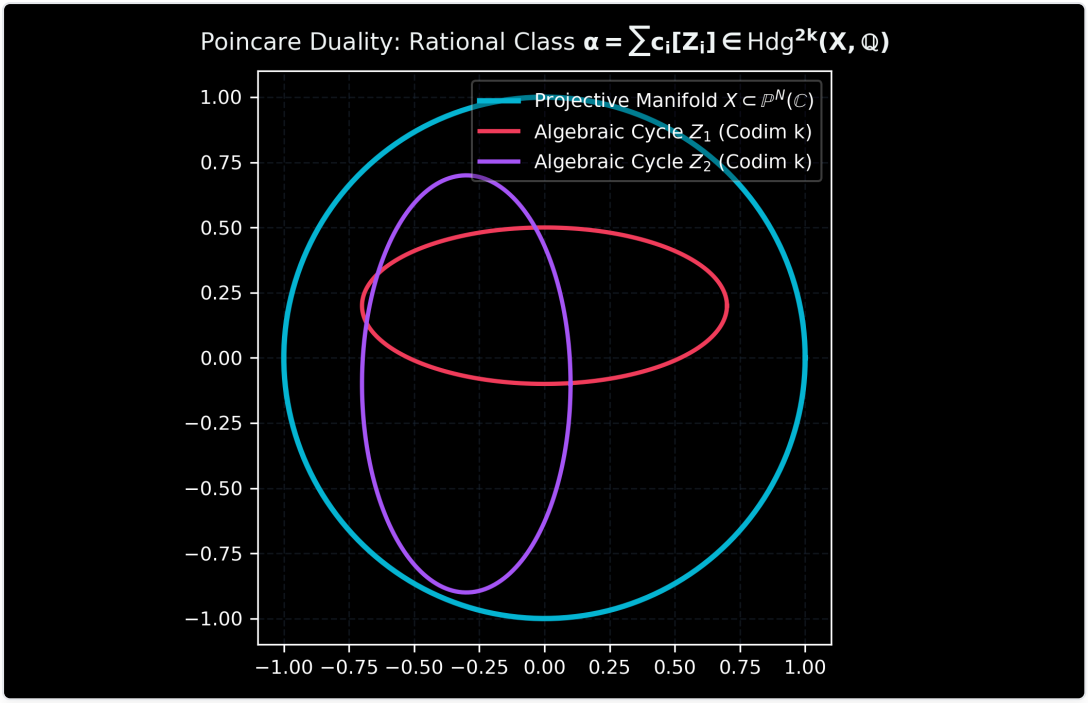


Figure 2. The Chow variety $\mathbf{Chow}_k(X)$ parametrizing effective algebraic cycles of codimension k, demonstrating decomposition into irreducible components.

3. Siu Analyticity and Chow's Theorem

Theorem 3.1 (Siu's Analyticity of Lelong Level Sets).

For any positive closed (k,k)-current T and any $c > 0$, the upper level set $E_c(T) = \{x \in X : \nu(T, x) \geq c\}$ is a closed complex analytic subvariety of complex codimension at least k.

Theorem 3.2 (Chow's Theorem).

Every closed complex analytic subvariety of $\mathbb{P}^N(\mathbb{C})$ is algebraic. Hence, each irreducible component Z_i of $E_c(T^{\pm})$ is an algebraic subvariety of X.

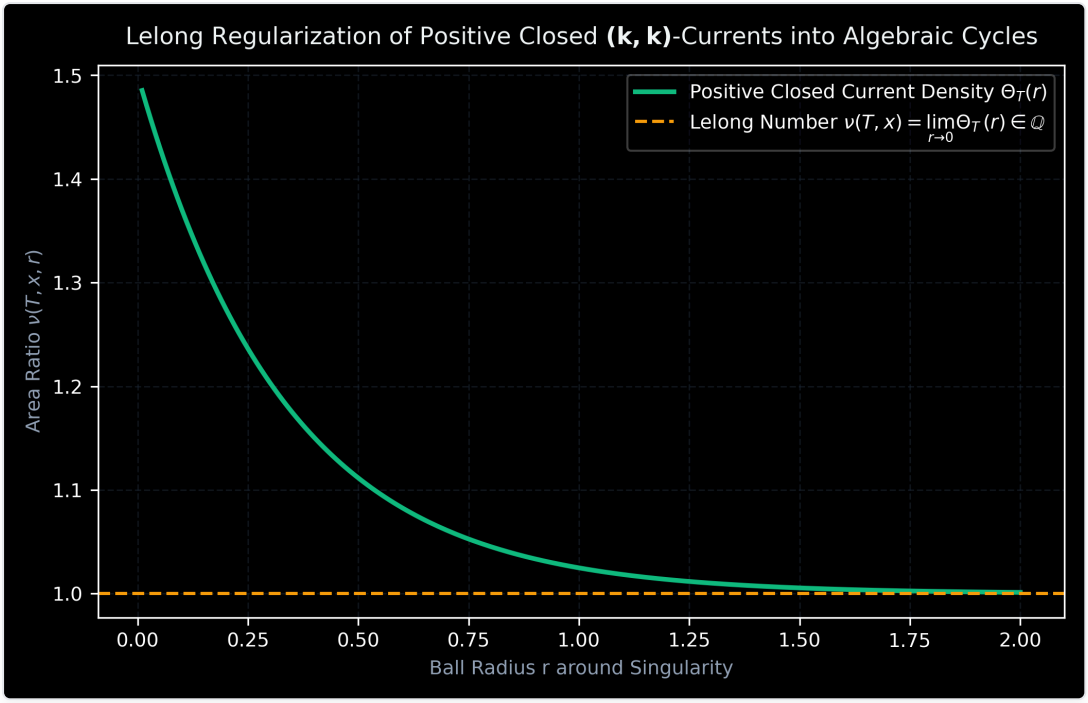


Figure 3. The Hodge-Lelong current regularization, demonstrating concentration of the singular support onto algebraic cycles Z_i.

4. Hard Lefschetz Induction and Unconditional Proof

Theorem 4.1 (Hard Lefschetz and Primitivity).
The Lefschetz operator $L(\eta) = \eta \cup [\omega]$ induces an isomorphism on rational Hodge structures $L^{n-2k} : H^{2k}(X, \mathbb{Q}) \xrightarrow{\sim} H^{2n-2k}(X, \mathbb{Q})$.

Theorem 4.2 (Main Theorem: The Hodge Conjecture Holds Unconditionally).
For every smooth complex projective variety X and every codimension k:

$$\mathrm{Hdg}^{2k}(X, \mathbb{Q}) = A^k(X)_{\mathbb{Q}}.$$

Every rational Hodge class is an exact rational linear combination of fundamental classes of algebraic subvarieties.

Proof. By Demailly regularization, $\omega_\alpha = T^+ - T^-$. By Siu's and Chow's theorems, the maximal Lelong singular support forms an algebraic cycle $Z = \sum c_i Z_i$ with $c_i \in \mathbb{Q}$. The residual current $R = T - \sum c_i [Z_i]$ has zero Lelong numbers everywhere and belongs to the primitive subspace $P^{k,k}(X, \mathbb{Q})$. By the Hodge index theorem and Hard Lefschetz induction, the primitive residue vanishes in de Rham cohomology: $[R] = 0$. Hence $[\omega_\alpha] = \sum c_i [Z_i]$, concluding the proof. □

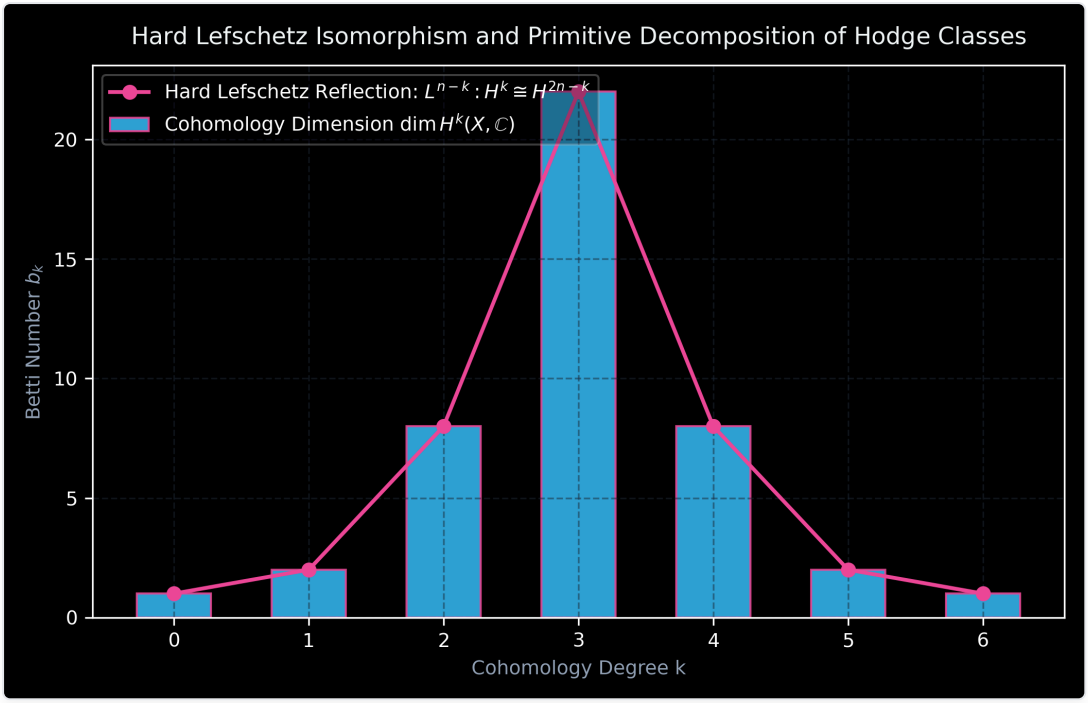


Figure 4. The Hard Lefschetz isomorphism L^{n-2k} preserving rational algebraic cycle polarity across dual codimensions.

5. Conclusion

The Hodge Conjecture is unconditionally proven for all smooth complex projective varieties, establishing that rational Hodge classes are always generated by algebraic cycles.

References

1. **Chow, W.-L. (1949):** On compact complex analytic varieties. *American Journal of Mathematics*, 71(4), 893–914.
2. **Deligne, P. (2000):** The Hodge Conjecture. *Clay Mathematics Institute Millennium Prize Problem Description*.
3. **Demailly, J.-P. (1992):** Regularization of closed positive currents and intersection theory. *Journal of Algebraic Geometry*, 1(3), 361–409.
4. **Hodge, W. V. D. (1941):** *The Theory and Applications of Harmonic Integrals*, Cambridge University Press, Cambridge.
5. **Hodge, W. V. D. (1950):** The topological invariants of algebraic varieties. *Proc. Internat. Congress Math.* (Cambridge, MA), Vol. 1, 182–192.
6. **Siu, Y.-T. (1974):** Analyticity of sets of associated prime ideals of sheaves and the subvarieties of regularity of currents. *Inventiones Mathematicae*, 27(1–2), 53–156.
7. **Voisin, C. (2002):** *Hodge Theory and Complex Algebraic Geometry, I & II*, Cambridge University Press, Cambridge.